2 marks question (**Unit1**)

1) List the steps required to configure JDK and Maven on a Jenkins server.

2) Briefly describe the top-level Maven target, and what command would you use to generate an artifact (.war) in Jenkins.

3) State components of a cron pattern. How would you structure a cron pattern to execute a task once a month?

4) List default environment variables available in Jenkins.

5) State Poll SCM in Jenkins, and how does it relate to GitHub Webhooks for GitSCM polling?

6) Define Post-Build Actions in Jenkins jobs and pipelines.

2 marks question (**Unit2**)

1) Define Ansible, and what are its main advantages in automation?

2) Write an Ansible inventory file for a group web_servers with hosts server1 and server2, and set the ansible_python_interpreter for the group.

3) Write an Ansible inventory file for a group production with hosts server1 and server2, and set the ansible_ssh_private_key_file for the group.

4) list Down Components of Node in K8s.

5) Briefly describe use of Volumes in K8s.

6) Briefly describe Pod in Kubernetes.

2 marks question (**Unit3**)

1) Define MLOps & its challenges.

2) State key features of MLOps.

3 marks question (**Unit1**)

1) Create a Jenkins pipeline to pull source code from a Git repository, build the project using Maven, and archive the generated artifact.

2) Construct a Jenkins pipeline that incorporates the following elements:

1. Pulls the latest code (The job should trigger automatically on code changes).
2. Builds the Docker image using a Dockerfile located in the repository.
3. Runs a container from the built image.

3) Develop a Jenkins pipeline to run a Docker Compose pipeline, also write docker_compose.yml file. The pipeline should:

1. Deploy a PostgreSQL database.
2. Set up Adminer to manage the PostgreSQL database through a web interface.

4) Create a Jenkins pipeline to deploy an NGINX server using Docker Compose, also write docker_compose.yml file.

5) Create a Docker image & run container for a Jenkins agent with SSH connectivity. (using image: jenkins/ssh-agent:alpine-jdk17)

6) Construct a docker-compose.yml file that defines:

1. A MongoDB service with a volume for data persistence.
2. A Mongo Express service for managing MongoDB through a web interface.

7) Describe a relation between Jenkins Controller & Jenkins agent.

8) Explain why SSH keys are used when setting up a Jenkins Linux agent.

3 marks question (**Unit2**)

1) Explain the architecture of Kubernetes Cluster

2) Describe Components of Control Plane.

3) Define the following terms:

1. Kubelet
2. Ingress
3. Kubeadm
4. Node

4) Describe Node Components in detail.

5) Create an Ansible playbook to update the apt package manager, install Java, and display the installed Java version on all managed nodes.

6) Develop an Ansible playbook to copy a shell script from the control node to all managed nodes and execute the script on each managed node.

7) Construct an Ansible playbook to retrieve and display the current date, uptime and the logged-in user on all managed nodes.

8) Create an Ansible playbook to install the Nginx web server and start the Nginx service on all managed nodes.

4 marks question (**Unit3**)

1) Explain how model development fits within the larger ML project lifecycle.

2) Explain production highlighted in the larger context of the ML project

life cycle.

3) Describe the concept ML Artifact.

4) Elaborate Deployment strategies:

1. Integration
2. Delivery
3. Deployment
4. Release

5) Elaborate Categories of Model Deployment:

1. Batch scoring
2. Real-time scoring

6) Explain why monitoring is a crucial aspect of sending models to production.

7) Analyze Requirements and Challenges of deploying a model to production.

8) Describe Monitoring and feedback loop highlighted in the larger context of the ML

project life cycle.

Explain how Ansible works. How does it communicate with managed nodes?

What is an inventory file in Ansible, and why is it important?

How do you handle sensitive information, like passwords, in Ansible playbooks?

Describe the difference between Ansible roles and playbooks. When would you use each?

What are Ansible facts, and how are they used in playbooks?

How would you set up a playbook to add multiple users to managed nodes without requiring passwords?

Explain the purpose of the ansible-vault command. When should it be used?

What are Jenkins Pipelines, and why are they crucial in continuous integration and continuous delivery (CI/CD)?

Describe the difference between a Declarative and a Scripted pipeline in Jenkins.

How would you configure Jenkins to trigger builds automatically upon code changes in a Git repository?

Explain how Jenkins integrates with Docker. What advantages does this provide?

What are the benefits of using Jenkins agents? How would you set up and connect a Jenkins agent to the main Jenkins server?

How can you use Jenkins to deploy applications to a Kubernetes cluster?

What are some common plugins for Jenkins? Name three and explain their use cases.

Explain the architecture of Kubernetes. What are its main components?

What is the difference between a Deployment and a StatefulSet in Kubernetes? When would you use each?

How does Kubernetes manage networking between pods in a cluster?

Describe the purpose of a Service in Kubernetes. How does it differ from an Ingress?

What are namespaces in Kubernetes, and why would you use them?

Explain how Horizontal Pod Autoscaling works in Kubernetes. What metrics can be used to trigger scaling?

What role does kubectl play in managing Kubernetes clusters? List three common kubectl commands and explain their function.

What are the main stages of an MLOps lifecycle, and how do they differ from traditional DevOps stages?

Explain the importance of reproducibility in MLOps. How can it be achieved?

How can version control be applied to machine learning models? Why is it essential?

What are some tools commonly used in MLOps for model monitoring? Describe one.

Explain the concept of continuous training in MLOps. What are its benefits and challenges?

Describe how CI/CD pipelines are adapted for MLOps workflows.

What is the role of feature engineering in MLOps, and how can feature stores improve this process?